

3. Methodology

The methodology followed in this work encompasses data collection, preprocessing, experimental setup, model selection, and performance evaluation. A complete description of each stage is provided below.

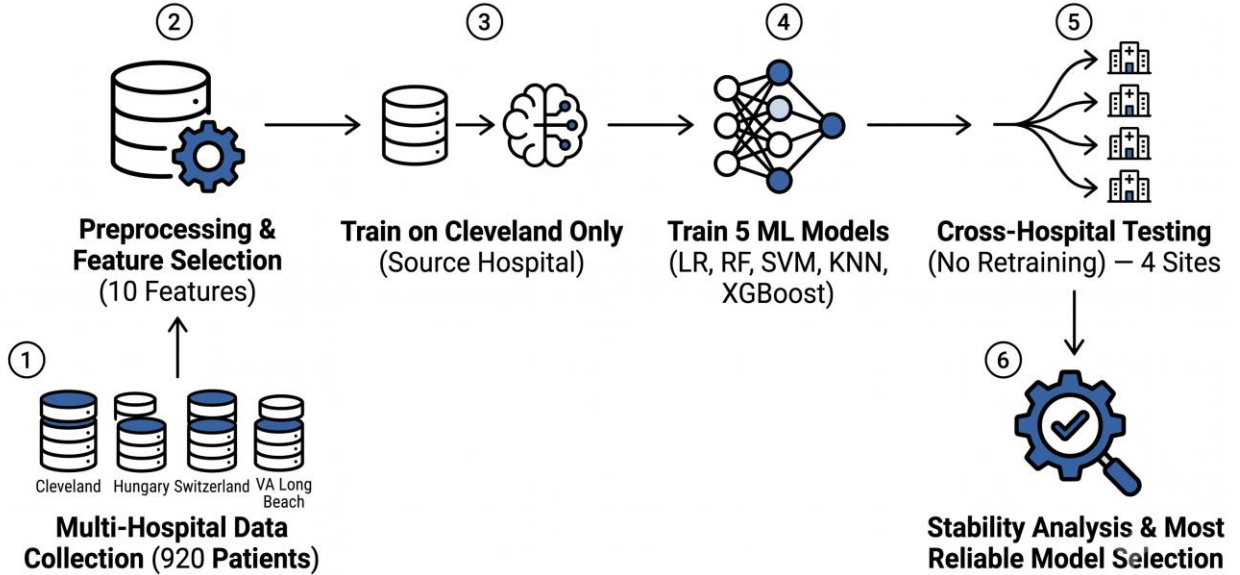


Figure 1: Cross-Hospital Generalization Architecture for Heart Disease Prediction

Figure 1 illustrates the complete workflow: multi-hospital data collection from four sites (Cleveland, Hungary, Switzerland, VA Long Beach), preprocessing and feature selection (10 common features), training exclusively on Cleveland (source hospital), training five machine learning models, cross-hospital testing on all four sites without retraining, and final stability analysis to identify the most reliable model.

3.1 Dataset Description

For our experiments, we used the well-established UCI Heart Disease repository. This dataset contains records from 920 patients drawn from four independent medical institutions: Cleveland, Hungary, Switzerland, and the VA Long Beach center. The dataset is publicly available and has been extensively utilized in predictive modeling studies.

We narrowed down the feature set to 10 variables that were consistently recorded across all four sites. These ten features are: age, sex, chest pain type, resting blood pressure, serum cholesterol,

fasting blood sugar, resting electrocardiographic results, maximum heart rate achieved, exercise-induced angina, and ST depression (oldpeak).

We simplified the original multiclass target into a binary label—0 for no disease and 1 for disease present. Several features were excluded from the analysis because they were largely missing in the Hungary, Switzerland, and VA Long Beach datasets, and imputing them would have introduced more guesswork than signal.

3.2 Data Preparation and Split Strategy

All data cleaning and preprocessing steps were performed using only the Cleveland partition. Missing numeric values were replaced with the median from Cleveland, and missing categorical values were replaced with the mode from Cleveland. This decision was critical to prevent information from the other three hospitals from influencing the training data.

For scaling, we applied standard normalization (zero mean, unit variance). The normalization parameters were learned exclusively from the Cleveland training portion and then applied to all other datasets. This simulates a realistic scenario in which a model is developed in one site and then transported to others.

The training set was formed by randomly selecting 80% of the Cleveland records. The remaining 20% of Cleveland, along with the complete datasets from Hungary, Switzerland, and VA Long Beach, were set aside as evaluation partitions.

3.3 Novelty of the Experimental Design

The key differentiator of our study lies in the evaluation scheme. Most earlier works have combined all sites. and performed random splits, we intentionally trained classifiers on a single site. This allows us to measure population-level generalizability, which is a more demanding and practically relevant test than within-sample accuracy. Our focus is not simply on predictive power, but on consistency across varying clinical populations.

3.4 Model Selection

We selected a set of five models, each representing different learning paradigm, to ensure a comprehensive comparison of model behavior across hospital populations. The chosen models include:

- Logistic Regression (linear classifier)
- Random Forest (bagged ensemble of trees)

- Support Vector Machine (SVM)
- K-Nearest Neighbors (distance-based lazy learner)
- XGBoost (boosted tree model)

All models were implemented using their default hyperparameters, with a fixed random seed (`random_state = 42`) applied throughout to ensure reproducibility. The one exception was XGBoost's learning rate, which was reduced from its default value of 0.3 to 0.1.

3.5 Evaluation Protocol

For evaluation of all models a range of standard classification metrics were computed to provide a comprehensive assessment of predictive ability of the models. This included:

- Accuracy – the proportion of correctly classified instances out of the total predictions.
- Precision – the ratio of true positive predictions to the total positive predictions, indicating how often the model is correct when it predicts disease.
- Recall (Sensitivity) – the ratio of true positive predictions to the actual positive instances, reflecting the model's ability to correctly identify patients with heart disease.
- F1 Score – the harmonic mean of precision and recall. It provides a balanced measure of performance, making it especially useful when both false positives and false negatives carry significant consequences, as is the case in medical prediction tasks

The F1 Score was prioritized due to its balance between false alarms and missed detections

To assess how consistently each model performs across different hospital populations, we introduced a stability metric called "Spread." Spread is defined as the range between the maximum and minimum F1 Score achieved by a model across the four hospitals.

Lower spread values indicate more consistent performance across heterogeneous populations, which we interpret as a proxy for methodological robustness rather than a claim of clinical readiness.